PMSCS PROGRAM

Department of Computer Science and Engineering
Jahangirnagar University

Date: 29/11/2024

Course Code: PMSCS-623P   **Title:** Data Structures and Algorithms

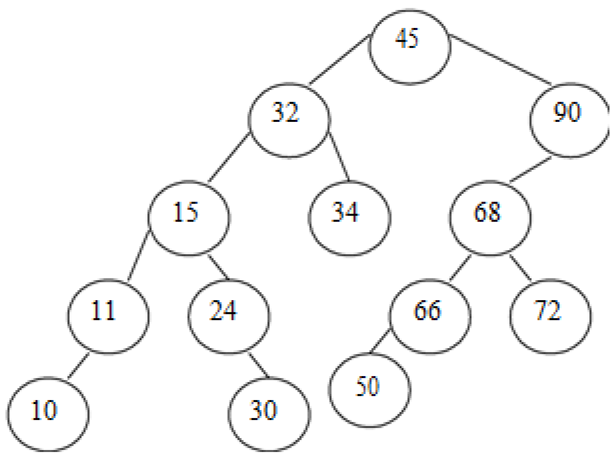
Time: 1.5 Hours   **TERM:** SUMMER 2024   **Full Marks:** 30

Answer any THREE from the following Questions

1.	a) Define *asymptotic* notation. Briefly explain *Theta* (Θ) notation.	**[1+3]**

	b) Analyze the running time of the following algorithm and write it using ***O***() notation:
	`int m = 10;`
	`for(int i=1; i <=m; i++){`
	`    for(int j=(2 × n); j>=1; j=j--){`
	`        printf("PMSCS 623P");`
	`    }`
	`}`
		[2]

	c) Perform Binary Search for the following array named *arr = {45, 77, 89, 90, 94, 99, 100}* and *key = 45*.	**[4]**
2.	a) What is *stack*? Transform the following infix expression into its equivalent postfix expression using *stack*: (a × b) – c) + (f × (d + e))	**[1+5]**
	b) Find the value of the following arithmetic expression written in *postfix* notation using *stack*: 12, P, 3, - , /, Y, 1, (Z+1) , + , × , + Where, P = alphabetic order of the 1st letter of your sur/last name Y = alphabetic order of the 2nd letter of your sur/last name Z = alphabetic order of the 3rd letter of your sur/last name **N.B.:** Make sure you write down your sur/last name at first and also the values of P, Y, Z.	**[4]**
3.	a) Mention the difference between *linear* and *non-linear* data structure with example.	**[2]**
	b) Do we consider actual running time for comparing algorithms? Justify your answer	**[3]**
	c) Briefly explain Divide and Conquer strategy. Perform **Mergesort** algorithm to an array {3, 22, 15, 8, 19, 10, 13, 8}.	**[1+4]**

4.	a) Find the value of the *postfix* expression: + / + 2 2 2 / – 3 2 + 1 0 P = alphabetic order of the 1^{st} letter of your first name Y = alphabetic order of the 2^{nd} letter of your first name Z = alphabetic order of the 3^{rd} letter of your first name **<u>N.B.:</u>** Make sure you write down your first name at first and also the values of P, Y, Z.	**[3]**
	b) Define Complete Binary tree with example.	**[2]**
	c) Traverse the following tree in Inorder and Postorder mode and write the vertices in sequence of visit in each case: 	**[5]**

N.B. Students must submit question along with answer scripts at the end of examination.